

Problem 11.11

In Fund X money accumulates at a force of interest

$$\delta_t = 0.01t + 0.1, \quad 0 \leq t \leq 20.$$

In Fund Y money accumulates at an annual effective interest rate i . An amount of 1 is invested in each fund for 20 years. The value of Fund X at the end of 20 years is equal to the value of Fund Y at the end of 20 years. Calculate the value of Fund Y at the end of 1.5 years.

Solution

The accumulated value of the unit deposit in Fund X at time 20 is

$$A_X(20) = \exp\left(\int_0^{20} \delta_t dt\right).$$

Substitute the given force of interest:

$$A_X(20) = \exp\left(\int_0^{20} (0.01t + 0.1) dt\right).$$

Evaluating the integral gives

$$\int_0^{20} (0.01t + 0.1) dt = [0.005t^2 + 0.1t]_0^{20} = 0.005(20)^2 + 0.1(20) = 4.$$

Therefore,

$$A_X(20) = e^4.$$

Fund Y accumulates at a constant annual effective interest rate i , so

$$A_Y(20) = (1 + i)^{20}.$$

Since the fund values at time 20 are equal,

$$(1 + i)^{20} = e^4.$$

Taking the 20th root,

$$1 + i = e^{4/20} = e^{0.2}.$$

Thus,

$$i = e^{0.2} - 1 \approx 0.2214027582.$$

The value of Fund Y at time 1.5 is

$$A_Y(1.5) = (1 + i)^{1.5} = (e^{0.2})^{1.5} = e^{0.3}.$$

Therefore,

$$\boxed{A_Y(1.5) \approx 1.349858808}.$$